

Digital twin aided adversarial transfer learning method for domain adaptation fault diagnosis

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ABSTRACT

Machine health management has become the focus of equipment monitoring upgrading with the advance of digital twin (DT). The DT model is able to generate system performance data that is close to reality, which opens a new way for the cyber-physical integration of equipment monitoring. Furthermore, it also provides a significant opportunity for mechanical fault diagnosis when the collected fault signals are insufficient. In this paper, a DT aided intelligent fault diagnosis model is proposed for triplex pump. Specifically, the simulation model of the triplex pump is built by Simscape in MATLAB, and the measured simulation data is continuously updated to construct the DT model. Then a novel transfer learning model based on domain-adversarial strategy and Wasserstein distance is present and trained by the source domain data which generated from the DT model. Next, the opening pressure of the triplex pump is controlled to simulate different working conditions, so as to achieve feature transfer and fault diagnosis for the DT model. The experimental results show that the proposed method is effective and superior to other advanced transfer learning methods.

1. Introduction

Industrial equipment always need to be forcibly shutdown when a fault occurs. The repair process may not cost much, but the economic loss will be significant due to the downtime [1,2]. Therefore, it is particularly important to carry out effective real-time online monitoring and fault diagnosis for mechanical equipment [3].

In recent years, intelligent fault diagnosis methods have made significant development with the rapid breakthrough of deep learning and advanced sensing technology [4,5]. Zhang et al. [6] present a renew fault diagnosis approach by using fast nonlinear blind deconvolution for mechanical fault diagnosis. Li et al. [7] developed a novel multisensory fusion method based on covariance matrix and multi-Riemannian kernel ridge regression to improve gearbox fault diagnosis performance. Wang et al. [8] established a subdomain adaptation transfer learning network to learn transferable features and achieved mechanical fault diagnosis in cross-domain field. Shen et al. [9] presented a hybrid robust convolutional autoencoder for machine tools anomaly detection. An et al. [10] constructed a self-learning transferable neural network to achieve self-believing, rectifying and doubting in mechanical fault prognostics. Even though these deep learning methods can achieve excellent

diagnosis performance, they still have severe data dependence. Meanwhile, collecting training data on such a large scale is quite expensive.

If the mechanical equipment itself can show the parts that are about to fail, and give a targeted maintenance strategy. Then it can save a lot of maintenance time to avoid economic losses caused by the downtime. In the development process of any intelligent fault diagnosis algorithm, the core is sensor data [11]. High-level features will be extracted from the collected sensor data during the training process of the fault diagnosis model, and these features will be used to test and diagnose the new data transmitted by mechanical equipment in real time [12]. However, it is impossible to obtain the sensor data from the actual mechanical equipment in some typical fault conditions [13,14]. If the mechanical equipment is destroyed artificially to obtain the corresponding data, it not only costs plenty of time and effort, but even may cause disastrous consequences.

The emergence and rapid development of digital twin (DT) theory provides a new way to solve the above problem [15,16]. The conceptual model of the DT is first proposed by Professor Grieves [17]. It can build an entire mapping virtual model of the mechanical equipment through the available high-fidelity physical model, historical sensor data and real-time data. So as to describe and reflect the full life cycle process of

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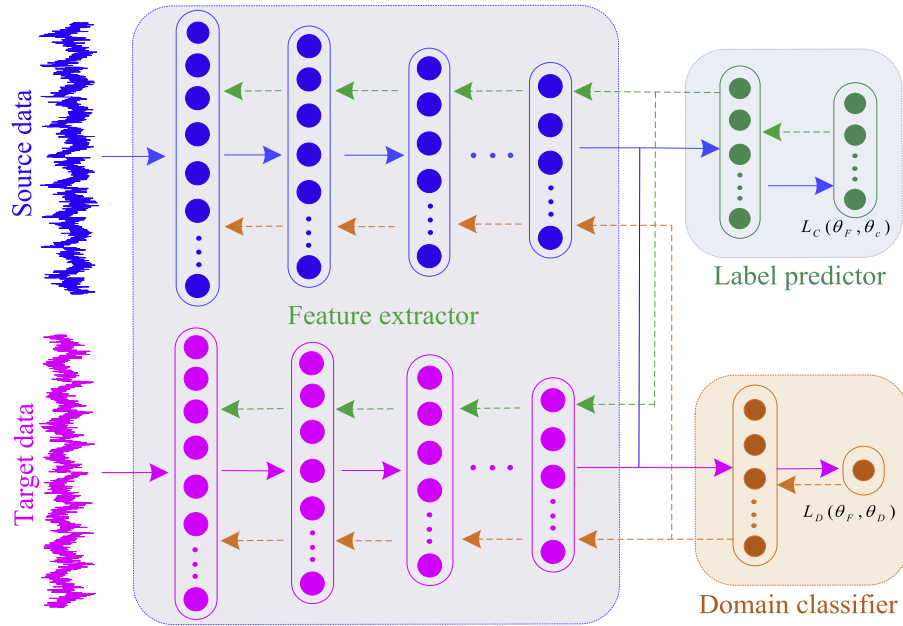


Fig. 1. Structure of DAAN.

the physical entity. Meanwhile, it can also generate simulation data for all sorts of fault situations [18]. Thus DT enables engineers to generate various sensor data needed for mechanical equipment health monitoring workflows, including measurable data of all possible failures with varying severity. Vanderhorn et al. [19] proposed a DT approach by combining vessel-specific operational data with computational models for the monitoring and prognosis of vessel-specific fatigue damage. Jain et al. [20] presented a DT model that enabled to real-time estimate the gagable feature output of a photovoltaic energy conversion unit (PVECU), and an error residual vector is generated and evaluated for the fault diagnosis process. Wang et al. [21] developed a DT model based on parameter sensitivity analysis for rotating machinery, and investigated its performance of unbalance quantification and localization for fault diagnosis. Rainer et al. [22] proposed a hydraulic circuit model based on DT and used it to generate simulated data for the training and development of the condition monitoring method. Xu et al. [23] present a two-phase digital-twin-assisted fault diagnosis method that can discover some potential problems at design time by pre-running the ultra-fidelity model in the virtual space. Shangguan et al. [24] proposed a novel physical-virtual converging DT system that integrates simulated data, real telemetry data and fusion data to achieve real-time monitoring and maintenance of on-orbit satellites.

In this paper, a novel DT aided adversarial transfer learning method for triplex pump intelligent fault diagnosis, so as to overcome the difficulty of collecting some real fault data. The main contributions of the paper can be summarized as follows:

- (1) A DT model of an actual triplex pump body is built and tuned to match the measurements in Simscape, so as to improve the reliability of the triplex pump health monitoring system.
- (2) A renewable DAAN method based on domain-adversarial strategy, Wasserstein distance and spectral normalization is constructed for domain adaption fault diagnosis.
- (3) A novel DT aided adversarial transfer learning method is proposed for intelligent fault diagnosis to ensure the safe operation of the triplex pump.

The remainder of this paper is organized as follows: Section II details the deep adaptive adversarial network (DAAN). Section III describes the proposed DT aided intelligent fault diagnosis framework. Section IV

presents the fault diagnosis experiment of the triplex pump. Section V finally provides conclusions, including:

2. Deep adaptive adversarial network

DAAN contains feature extractor, label predictor and domain classifier based on domain-adversarial training of neural networks [25] as displayed in Fig. 1. The feature extractor can extract fault features automatically from source and target domains, and label predictor classifies fault types relies on the learned features. The domain classifier is completed by a discriminator network based on domain adversarial strategy to narrow the distribution discrepancy between different domains.

2.1. Label predictor

The optimization object is constructed to train the feature extractor and label classifier C with parameter θ_F and θ_C . The loss function L_P is defined by the softmax cross-entropy between predicted probabilistic distribution and label data as follows:

$$L_P(x^s, y^s) = -\frac{1}{n^s} \sum_{i=1}^{n^s} \sum_{k=1}^K l(y_i^s = k) \cdot \log C(F(x_i^s))_k \quad (1)$$

where K denotes the number of health conditions, $l(y_i^s = k)$ denotes the indicator function, and $C(F(x_i^s))_k$ denotes the k th value of the predicted distribution.

2.2. Domain classifier

The adversarial training strategy is adopted to extract domain invariant features from the source and target domains. The optimization process of the domain classifier is to maximize the domain adversarial loss L_D and minimize the distribution discrepancy:

$$L_D = \frac{1}{n^s} \sum_{i=1}^{n^s} E_{x^s \sim p_s} [D(F(x_i^s))] + \frac{1}{n^t} \sum_{j=1}^{n^t} E_{x^t \sim p_t} [1 - D(F(x_j^t))] \quad (2)$$

Combining the above two optimization objects, then the final cost function is obtained as follows:

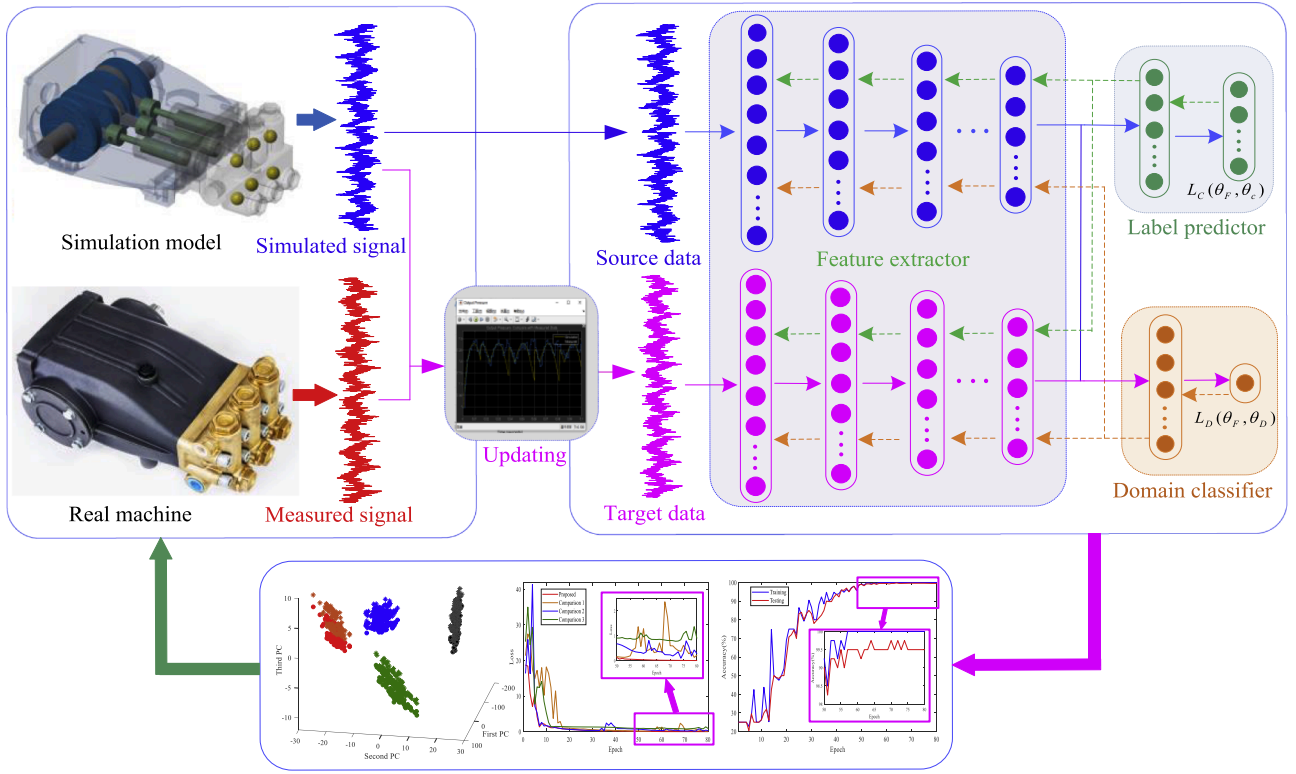


Fig. 2. Framework of digital twin aided intelligent fault diagnosis.

$$L = L_p + \lambda L_D \quad (3)$$

where the hyperparameter λ is used to determine the domain adversarial strength.

Wasserstein distance [26] is an effective distance metric that used to compare probability measure and distribution. In DAAN, it is adopted in domain classifier D as the distribution measure function to reduce the problem of gradient vanishing, which is adopted as the minimum cost to converge p_g to p_r :

$$W(p_r, p_g) = \inf_{\gamma \in \Pi(p_r, p_g)} E_{(x,y) \sim \gamma} [\|x - y\|] \quad (4)$$

where $\gamma(x, y)$ denotes the cost of moving an amount from x to y so as to transform p_r into p_g , $\Pi(p_r, p_g)$ denotes all joint distribution set $\gamma(x, y)$ whose marginals are p_r and p_g , respectively. $W(p_r, p_g)$ denotes the minimum transportation cost under optimal path planning.

Therefore, the enhanced cost function is achieved as follows:

$$\min_G \max_{D \in \mathbb{R}} E_{x \sim p_r} [D(x)] - E_{y \sim p_g} [D(y)] \quad (5)$$

where $\mathbb{R}$ denotes the 1-Lipschitz function set.

Spectral normalization [27] is applied to D to stabilize the training process. It can control the Lipschitz constant of domain classifier function by strictly limiting the spectral norm of each layer, so that the

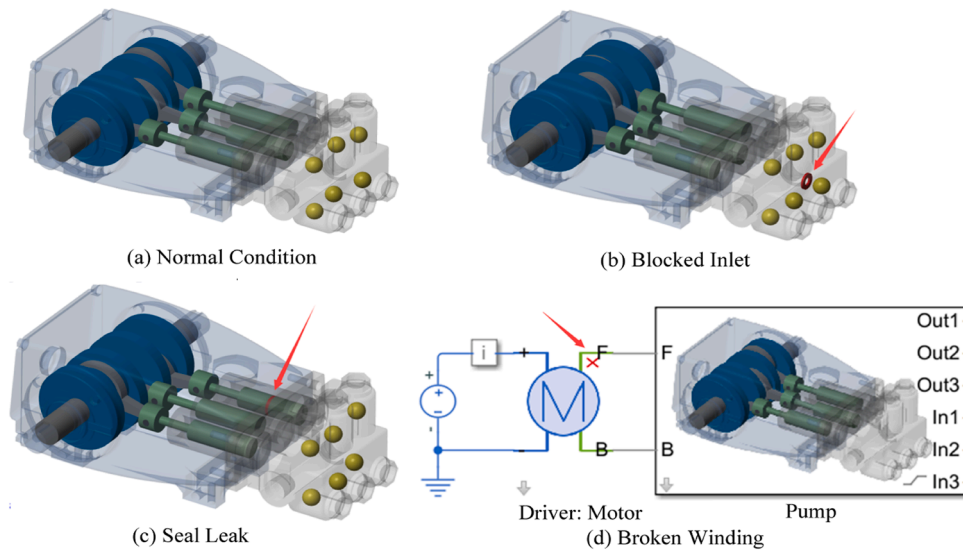


Fig. 3. Health conditions of triplex pump.

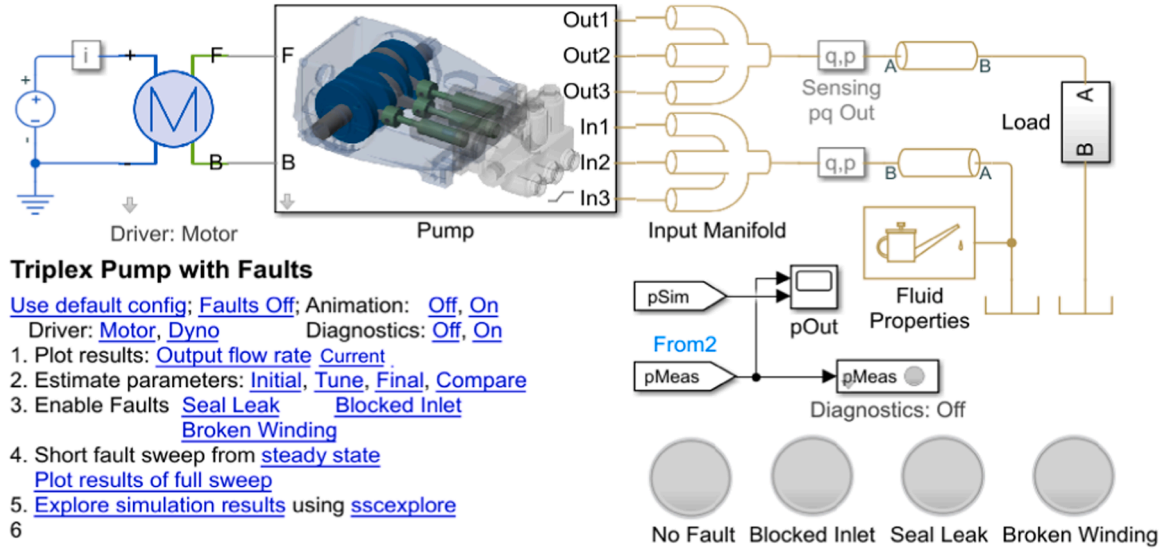


Fig. 4. Simscape of triplex pump.

domain classifier does not make intensive adjustment when the Lipschitz constant is the only hyper-parameter. Adam algorithm [28] is used to train the optimization object of the domain adversarial network. In the domain classifier, a gradient reversal layer [29] is added to connect the feature extractor, which can make sure the feature distribution of different domains is already indistinguishable enough to extract domain invariant features. So the loss is rewritten as follows:

$$L(\theta_F, \theta_C, \theta_D) = \min_{\theta_F, \theta_C, \theta_D} (L_C(\theta_F, \theta_C) + \lambda L_D(\theta_F, \theta_D)) \quad (6)$$

Then, the θ_F , θ_C , and θ_D are updated as follows:

$$\theta_F \leftarrow \theta_F - \alpha \left(\frac{\partial L_C}{\partial \theta_F} + \lambda \frac{\partial L_D}{\partial \theta_F} \right) \quad (7)$$

$$\theta_C \leftarrow \theta_C - \alpha \frac{\partial L_C}{\partial \theta_C} \quad (8)$$

$$\theta_D \leftarrow \theta_D - \alpha \frac{\partial L_D}{\partial \theta_D} \quad (9)$$

where α denotes the learning rate.

3. Digital twin aided intelligent fault diagnosis framework

The proposed digital twin aided intelligent fault diagnosis framework for triplex pump is displayed in Fig. 2. The CAD model of the triplex pump can be obtained from the manufacturer and is able to be imported into Simulink to create mechanical model for 3D simulation [30]. The triplex pump has a crankshaft to drive three plungers as shown in Fig. 3(a). One chamber of the plunger is always vented compared to a single piston pump, making the flow smoother and reducing pressure changes. The DT model is employed to generate simulated data of some common triplex pump health conditions. Then the simulated data is used as source domain data of the DAAN mode, and the updating data is used as target domain data. Finally, the trained model after transfer learning can be used for triplex pump fault diagnosis. The detailed fault diagnosis procedure is presented as follows:

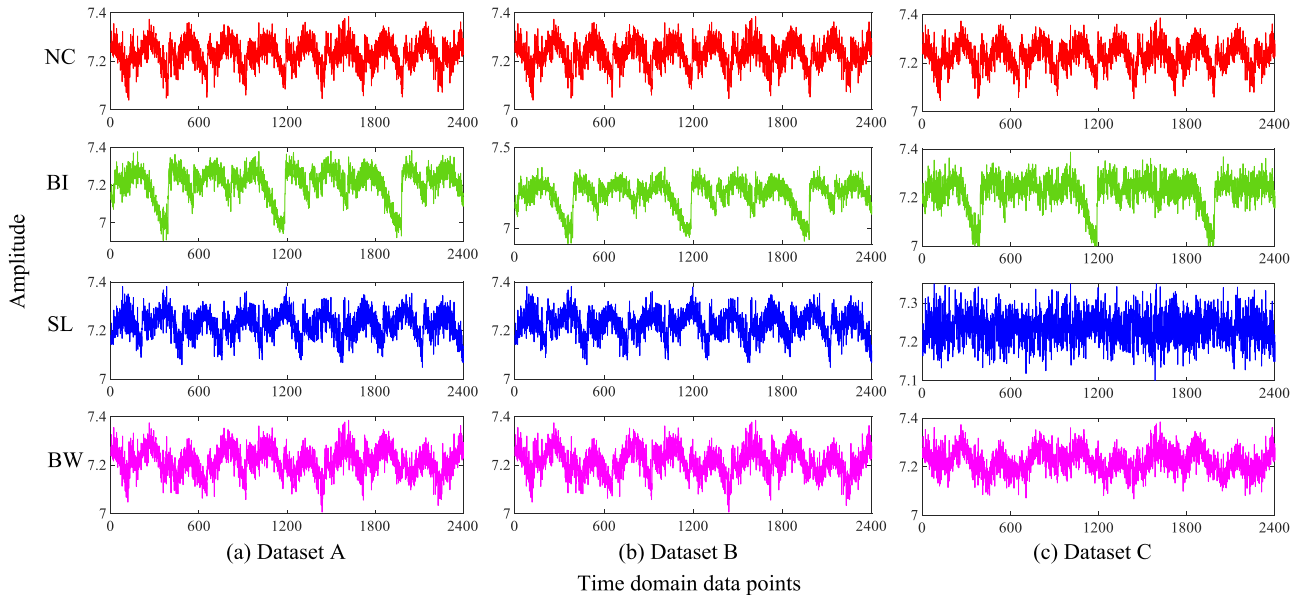


Fig. 5. Health conditions of triplex pump driven by dyno motor.

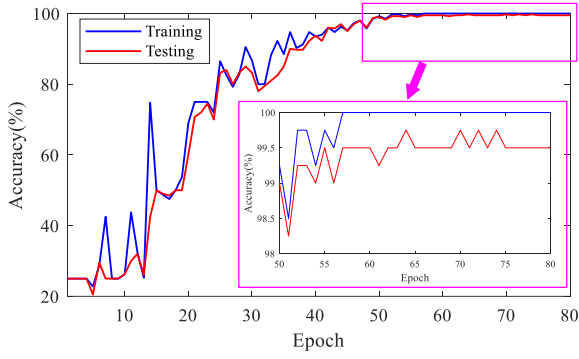


Fig. 6. Changes in training and testing accuracies of DAAN.

- 1) Setting different fault scenarios of the triplex pump in Simscape.
- 2) Collecting simulated outlet pressure signals of triplex pump under different fault scenarios.
- 3) Updating the model by averaging method to minimize the difference between the simulated data and measured data.
- 4) Construting the transfer learning model based on DAAN, the simulated data and the updating data are used as source domain and target domain data to pre-train the DAAN model.
- 5) Achieving intelligent fault diagnosis of the triplex pump by model training and testing.

4. Experimental verification

4.1. DT model of triplex pump driven by electro motor

The designed faults of this pump are blocked inlet (BI), seal leak (SL), and broken winding (BW) as shown in Fig. 3(b)–(d). To simulate the dynamic behavior of the system, the triplex pump is supplemented with hydraulic and electric components as shown in Fig. 4. The reciprocating pump consists of an electric motor, the pump housing, pump crank and pump plungers. The pump is driven by an electric motor, and the sensor is adhered to the outlet of the pump to collect the pressure data. The maximum opening pressure of the pump is $1.40E + 5$ Pa and the sampling period is $7.812E-5$ kHz. Simulink Design Optimization (SDO) is used to automatically tune parameter values so that the model generates results that match the measured data. The selected optimization parameters are shown in the check valve outlet module in Simscape. SDO selected parameter values for simulation, and calculated the difference between the simulation curve and the measured curve. Then, a new parameter value is selected for a new simulation based on the results. So as to calculate the gradient of parameter value and determine the direction of parameter adjustment. In this case, the convergence is fast because only two parameters are tuned. It is important to use features that speed up the tuning process for more complex scenarios with more parameters.

Three datasets marked as Dataset A, B, and C are collected to simulate three different scenarios. Dataset A represents the simulated data from the simulation model. Dataset B represents the actual scenario data by changing the maximum opening pressure to $1.38E + 5$ Pa, which is restricted by the experimental condition. Dataset C represents the updating data that optimized by the DT model. The simulation duration for each run is 20s. Each health condition includes 100 samples, and each sample includes 2400 data points as shown in Fig. 5.

The spectra of all the samples are used as input to train the DAAN model, the structure of the label predictor network is [1200, 600, 200, 100, 5], and the domain classifier network is [1200, 600, 200, 100, 1]. The learning rate is 0.005, batch size is 400, and penalty parameter λ is 0.02. 15 trials were conducted in each experiment to reduce the effect of the randomness, and the training epoch is 80. Dataset A is used as source

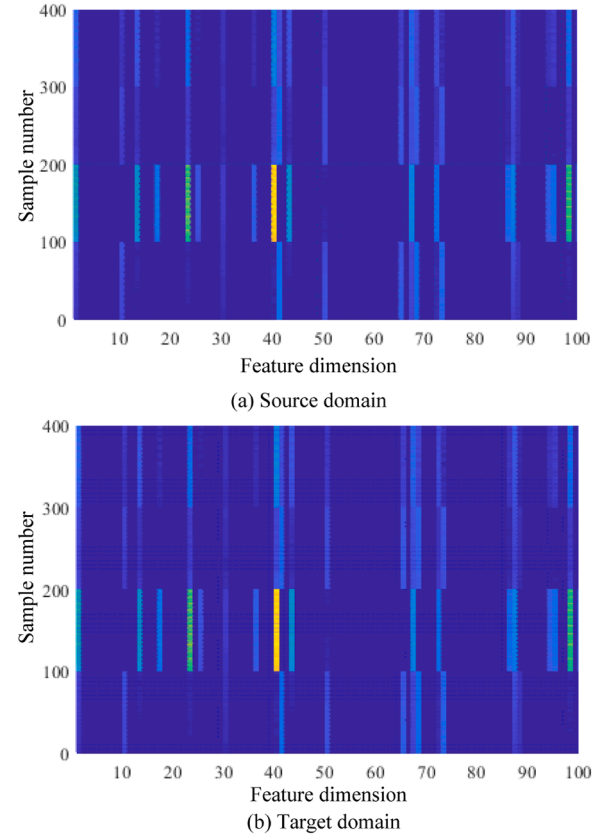


Fig. 7. Learned features of DAAN.

Table 1
Comparison of training and testing accuracies in Case 1.

Method	Proposed	Comparison 1	Comparison 2	Comparison 3
Training accuracy	100%	100%	96.44% $\pm$ 0.95%	86.57% $\pm$ 2.51%
Testing accuracy	99.50% $\pm$ 0.25%	95.42% $\pm$ 1.46%	88.68% $\pm$ 2.46%	75.12% $\pm$ 3.57%

domain data, Dataset C is used as target domain data, and Dataset B is used for testing. The curves of training and testing accuracies are displayed in Fig. 6. It can be seen that the training accuracy has reached 100% after approximately 57th epoch, and the testing accuracy is stabilized at $99.50\% \pm 0.25\%$ after 80th epoch. The learned domain invariant features of source and target domains are displayed in Fig. 7, it is observed that all the feature vectors of same health condition show almost one feature trend, and the feature vectors in different domains also exhibit consistency. It indicates that the DT aided intelligent fault diagnosis method can effectively extract domain invariant features and accurately classify the health condition of triple pump.

To demonstrate the effectiveness of the proposed method, three comparison method are employed:

- 1) Comparison 1: Dataset B is used as target domain data to participate in training.
- 2) Comparison 2: Hybrid distance-guided adversarial network (HDAN) [31].
- 3) Comparison 3: Distance guided domain-adversarial network (DGDAN) [32].

Comparison 1 is used to evaluate the superiority of the updating data by the proposed method, comparison 2 and 3 are employed to evaluate

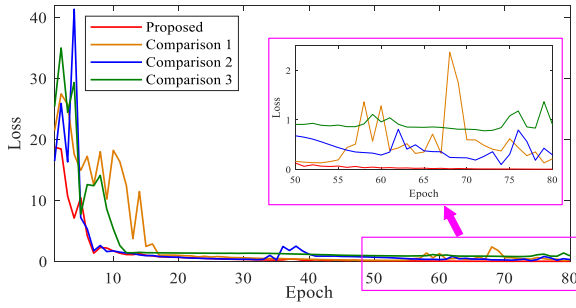


Fig. 8. Comparison of training losses.

the effectiveness of the transfer learning method based on DAAN. The comparison result of training and testing accuracies are displayed in Table 1. It is obvious that the proposed method has obtained the highest training and testing accuracies than the other method. Comparison 1 has also achieved 100% training accuracy, but the testing accuracy is just $95.42\% \pm 1.46\%$. Thus it can be seen that the updating Dataset C is able to proceed more stable transfer learning training than Dataset B. In addition, Comparison 2 and 3 are two traditional transfer learning methods that using the same datasets as the proposed method for training. Their training accuracies are all not achieved 100%, and the testing accuracies are $88.68\% \pm 2.46\%$ and $75.12\% \pm 3.57\%$ that are quite lower than the proposed method. Furthermore, the classifier loss curves of the above methods are plotted in Fig. 8. It is observed that the loss curve of DAAN converges to zero after approximately 70th epoch, and the other methods are all not performing convergence until the ending of the training. Therefore, it can be concluded from the above

comparisons that the proposed method can conduct more effective transfer learning training and extract more robust domain invariant features than the other methods.

To further prove the effectiveness of the proposed method on fault classification, Principal Component Analysis (PCA) [33] is used to map the extracted high dimensional features into a 3-D scatter diagram for distribution visualization on the two domains. The features come from the last layer of the feature extractor (the pink one in Fig. 2) that are all from the test set. The visualization results of all the methods are displayed in Fig. 9. S and T are used to denote the source domain and target domain, respectively. The result in Fig. 9(a) shows that the proposed method not only can clearly classify different health conditions, but also can reduce the distribution discrepancy of the two domains. The visualization result of Comparison 1 in Fig. 9(b) is inferior to the proposed method. Even though the distribution discrepancy is quite well, the classify results of SL and BW are mixed together. The situation of Comparison 2 is similar to Comparison 1 that SL and BW are also not clustered well as shown in Fig. 9(c). Fig. 9(d) displays that the learned features via Comparison 3 have not performed obvious distinction, and all the samples except BI are mixed with each other. Consequently, the proposed method is able to effectively reduce the discrepancy between different domains and well classify all the health conditions of triplex pump.

4.2. DT model of triplex pump driven by dyno motor

In this section, another experiment of the triplex pump driven by dyno motor is designed as shown in Fig. 10. A new fault type of worn bearing is set as shown in Fig. 11, since the DC motor can not simulate the BW fault. The sampling period is 1E-3 kHz. The simulation duration

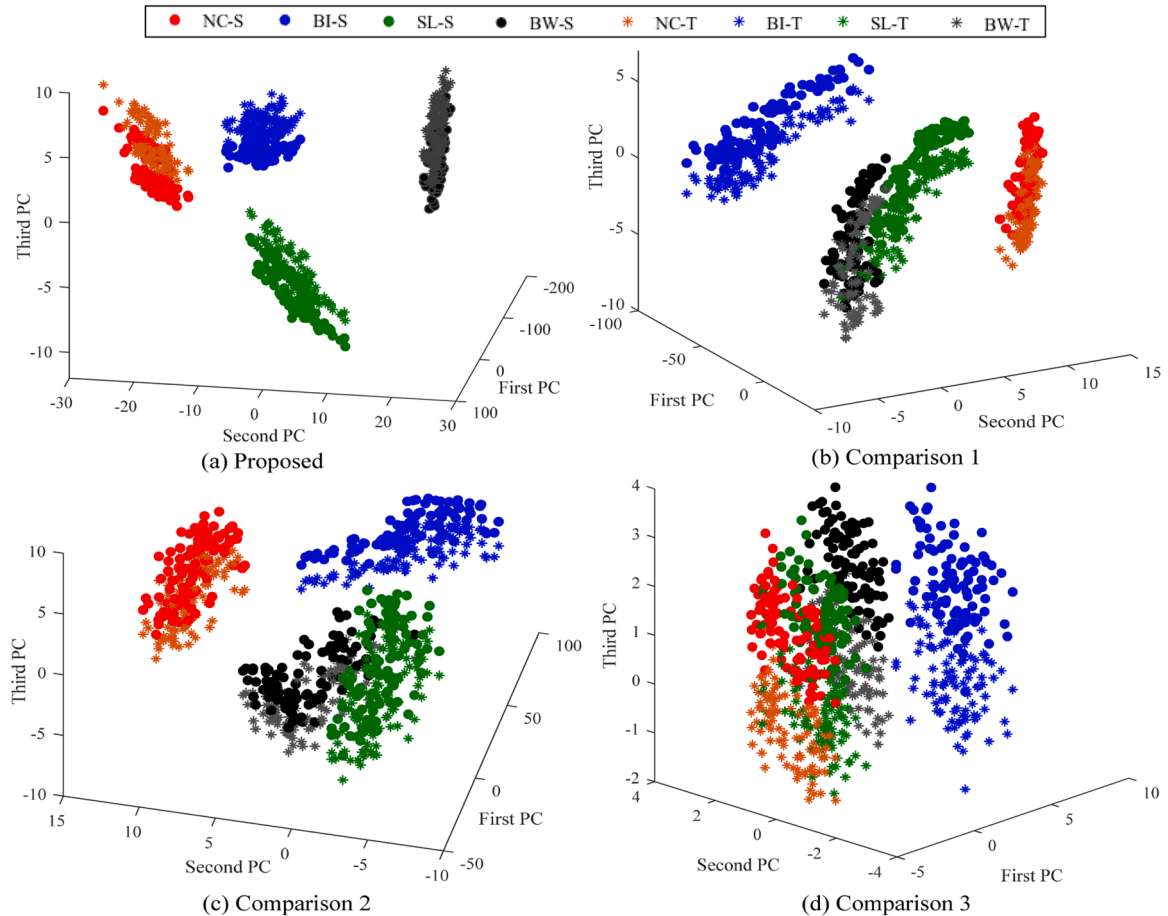


Fig. 9. Scatter diagram of dimension reduction results in case 1: (a) Proposed, (b) Comparison 1, (c) Comparison 2, and (d) Comparison 3.

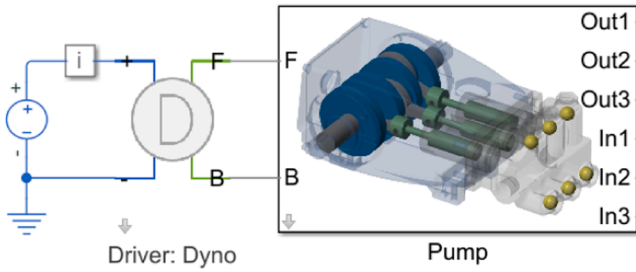


Fig. 10. Simscape of triplex pump driven by dyno motor.

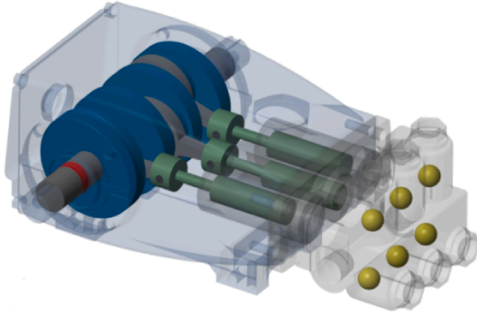


Fig. 11. Worn bearing fault type.

for each run is 40s. Each health condition includes 100 samples, and each sample includes 1200 data points as shown in Fig. 12(a). The spectra of all the samples are also used as input to train the DAAN model, the structure of the label predictor network is [600, 200, 100, 5], and the domain classifier network is [600, 200, 100, 1]. The other parameters are the same as in Case 1. There are also three datasets as shown in Fig. 12 in this case study. Dataset D is the simulated data that used as source domain data, Dataset F is another working condition data that used as target domain data, and Dataset E is the updating data that used for testing.

20 trials are also carried out to test the effectiveness of the proposed method and the comparison method, and the diagnosis results are displayed in Table 2. It is observed that the proposed method can also achieve the highest accuracy (100% training accuracy and 98.78% ±

0.34% testing accuracy). The comparison 1 method can also obtain 100 training method, but the testing accuracy is 93.25% ± 1.98% which is lower than the proposed method. The other two comparison methods can not achieve so high diagnosis accuracies no matter the training and testing processes. Therefore, it is found that the proposed method is also effectiveness in the other working condition of the triplex pump.

In addition, PCA is also employed to map the learned high dimension features of all the methods into 3-D scatter diagrams for distribution visualization. The dimension reduction results are exhibited in Fig. 13. We can clearly see that the proposed method also achieves the best dimension reduction result as shown in Fig. 13(a) that the different domain samples are clustered together, and different health condition samples are separated with each other. Fig. 13(b)–(d) show the dimension reduction results of all the three comparison methods. It can be seen that the result of Comparison 1 is also quite well, but the clustering effect of SL is relatively poor as shown in Fig. 13(b). The performance of Comparison 2 is worse since the samples of SL and BI in different domains are completely separated as shown in Fig. 13(c). Fig. 13(d) displays the performance of Comparison 3, it is clearly seen that three health condition samples are all mixed with each other that show no clustering effect. By comparison of above various results, it is obvious that the proposed method can also achieve the highest diagnosis and the best classify result than the other comparison method.

5. Discussion

In the field of intelligent fault diagnosis, data-driven methods almost have no connection to the virtual model of machinery. They always become inapplicable when the collected data are insufficient or the working condition is abrupt. DT technology can build a relationship between the virtual model and physical device, it can generate simulation data to describe the behavior of the real machinery and achieve domain adaptive fault diagnosis. The main advantage of our method is

Table 2
Comparison of training and testing accuracies in Case 2.

Method	Proposed	Comparison 1	Comparison 2	Comparison 3
Training accuracy	100%	100%	95.23% ± 1.20%	83.67% ± 2.47%
Testing accuracy	98.78% ± 0.34%	93.25% ± 1.98%	85.42% ± 3.54%	71.52% ± 3.41%

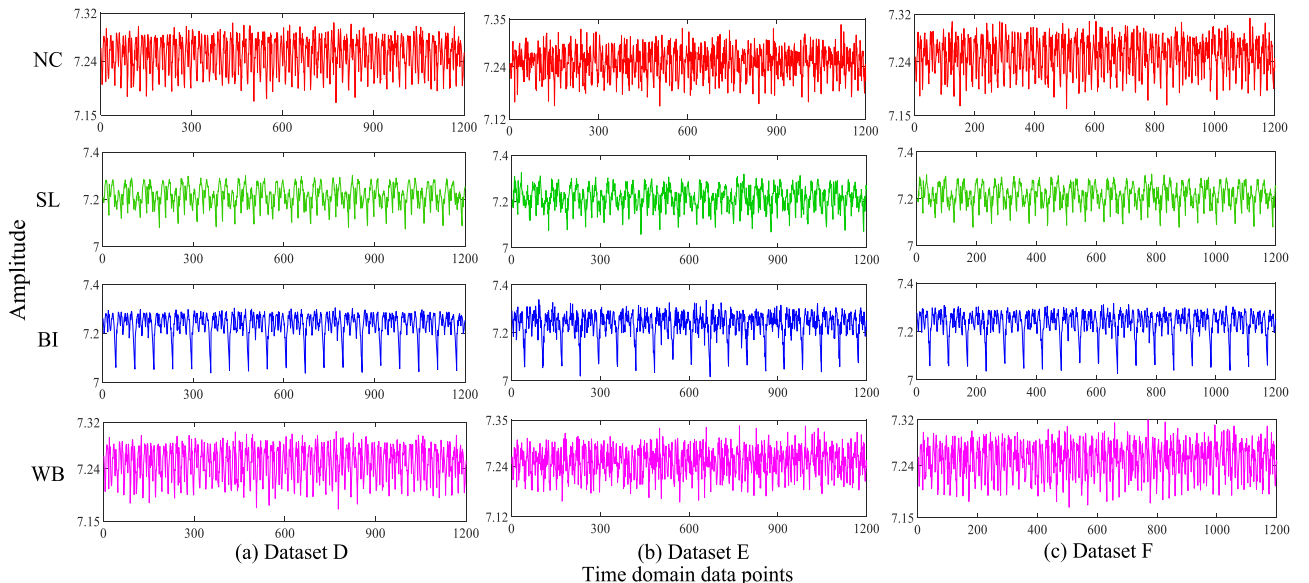


Fig. 12. Health conditions of triplex pump driven by dyno motor.

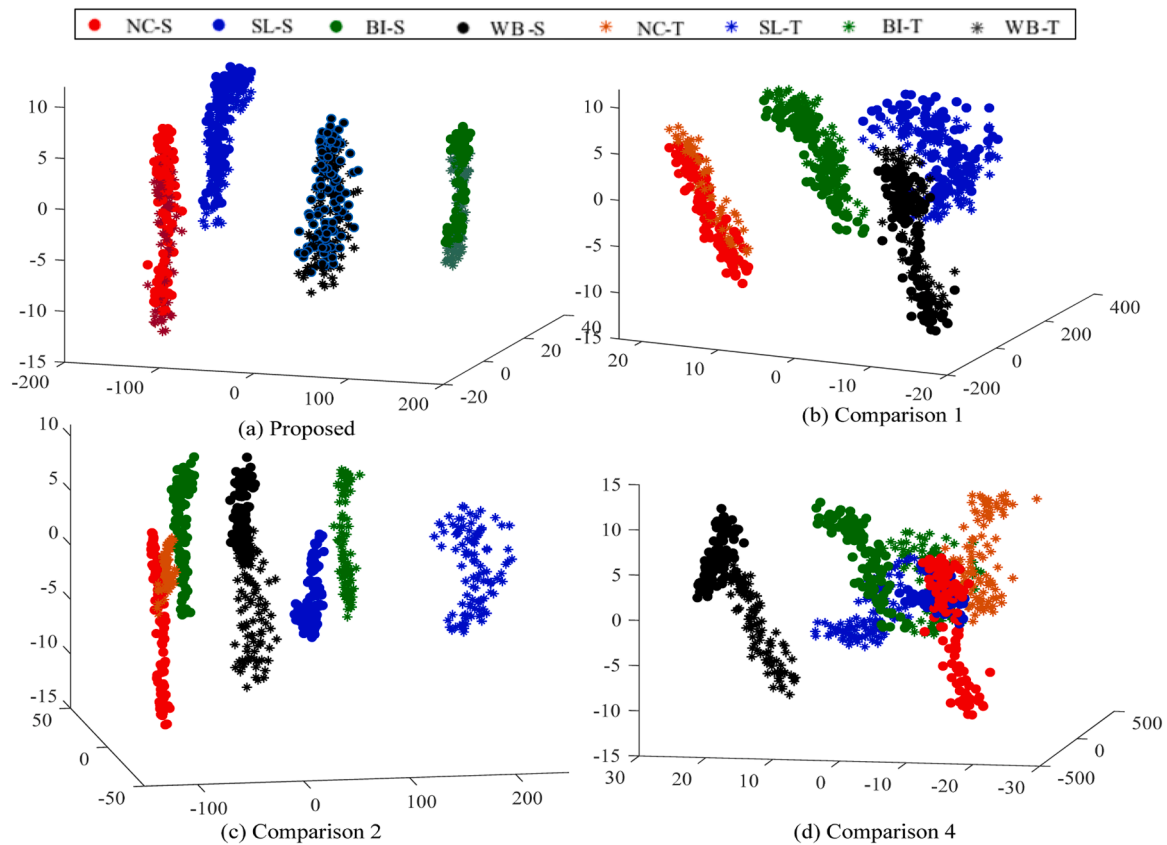


Fig. 13. Scatter diagram of dimension reduction results in case 2: (a) Proposed, (b) Comparison 1, (c) Comparison 2, and (d) Comparison 3.

to establish an executable DT aided adversarial transfer learning method for domain adaptation fault diagnosis of triplex pump. It can achieve an integration of the simulation model and the real device so as to enhance the performance of the domain adaptation fault diagnosis. The simulated model is synchronized with the real device based on the data interactive mapping strategy, which can be updated to diagnosis the triplex pump health condition. It provides an effective way to improve the reliability of the health monitoring system and ensure the safe operation of the triplex pump.

However, there are still some deficiencies needing to be addressed. The first challenge is to construct a more comprehensive DT model for the monitoring and operation of the triplex pump, since the proposed model has not consider some reality working conditions and the evolution process of fault occurrence. In addition, it requires the development of more advanced transfer learning algorithms in the DT framework and the extension of the method to other machines.

6. Conclusions

This paper proposed a digital twin aided adversarial transfer learning method for triplex pump fault diagnosis. Simscape is used to construct a digital twin model of the triplex pump, and the model is constantly updated when the characteristics of the pump changed. Simulated fault data were generated from the digital twin model to constitute the source domain data, and the updated data were used to constitute the target domain data. Then a novel domain adaptive adversarial network is present to learn domain invariant features and achieve intelligent fault diagnosis of the triplex pump. The effectiveness of the proposed method was verified using generated simulation data from different working scenarios of the triplex pump. The experimental results showed the superiority of the proposed method on feature transfer and fault classification than the other methods. In the future, more actual measured

dataset will be used to verify and enhance the proposed method.

CRediT authorship contribution statement

Jinrui Wang: Methodology, Writing – original draft, Funding acquisition. **Zongzhen Zhang:** Data curation, Software. **Zhiliang Liu:** Formal analysis, Investigation. **Baokun Han:** Methodology, Supervision. **Huaiqian Bao:** Formal analysis, Conceptualization. **Shanshan Ji:** Writing – review & editing, Software.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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